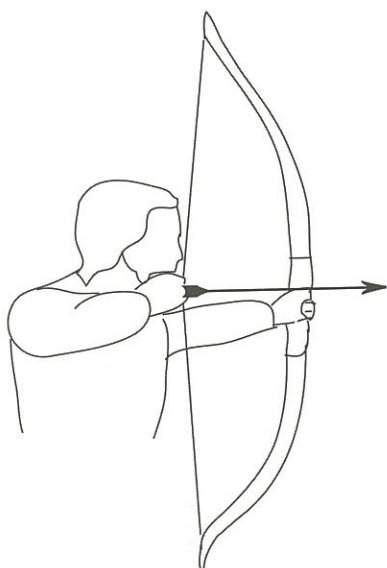


# Problem Solving and Calculations

## Set 1: Vector addition, subtraction and resolution

### Notes



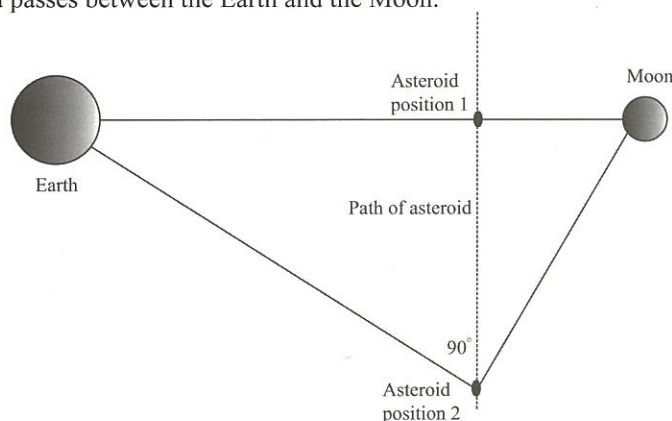
1. Explain why it is essential to describe both magnitude and direction of a quantity such as velocity.
2. A hockey player runs from fullback directly up the field for 15 m in an easterly direction to intercept the ball. He then dribbles the ball for 10 m in a northerly direction before passing the ball to a team-mate 20 m to the west.
  - a) Calculate the player's displacement.
  - b) Calculate the ball's displacement.
3. An archer stretches a bow so the string makes angles of  $45^\circ$  to the arrow. If the tension in the string is 208 N, what force is exerted on the arrow at the time of release?
4. A swimmer can achieve a speed of  $1.5 \text{ m s}^{-1}$  in still water. She heads directly across a rip in which the water is moving at a velocity of  $3.5 \text{ m s}^{-1}$  west. Determine her resultant velocity.
5. A canoeist can paddle a slalom kayak for short bursts at a speed of  $2.7 \text{ m s}^{-1}$  in still water. He wants to cross a stream in which a  $2.0 \text{ m s}^{-1}$  current flows.
  - a) At what angle to the current must he point his canoe if he wants to land on the other bank directly opposite to where he started? Assume he paddles at top speed throughout his crossing.
  - b) Calculate his resultant velocity if he heads directly across the stream at top speed.
  - c) The stream is 40 m wide. How far downstream will he land on the opposite bank?
6. A cross country skier uses a compass to determine direction. She leaves a ski village and skis 3.0 km east, then 8.0 km south, then 10.0 km east, then 5.0 km south, then 3.0 km west, then 6.0 km north, and finally 4.0 km west. She wants to return to village by the most direct route.
  - a) Draw a diagram to an appropriate scale and work out how far from the village she is.
  - b) In what direction should she head to return to the village if the terrain will allow her to travel straight to the village?
7. A netball player is standing still when she catches the ball, that is moving at  $5.5 \text{ m s}^{-1}$ . Determine the ball's change in velocity.
8. A tennis player serves a ball at  $80 \text{ km h}^{-1}$  towards his opponent, who returns it directly back to the server at  $90 \text{ km h}^{-1}$ . Calculate the ball's change in velocity.
9. In a game of squash a player strikes the ball so that hits the side wall at  $25 \text{ m s}^{-1}$  at an angle of  $45^\circ$  to the wall. It rebounds at  $20 \text{ m s}^{-1}$  at an angle to the wall of  $45^\circ$  on the opposite side of the normal. Calculate the ball's change in velocity.

# Problem Solving and Calculations 1

## Set 1: Vector addition, subtraction and resolution

### Notes

10. Two ice skaters hold one arm each of a third skate and each pulls with a force of 150 N. Calculate the resultant force pulling the third ice skater forward if the angle between his arms is a right angle.
11. A batter hits a cricket ball at right angles to its original direction. The bowler bowled the ball at  $30 \text{ m s}^{-1}$ . If the ball leaves the bat at  $30 \text{ m s}^{-1}$ , determine the ball's change in velocity.
12. A small asteroid passes between the Earth and the Moon.

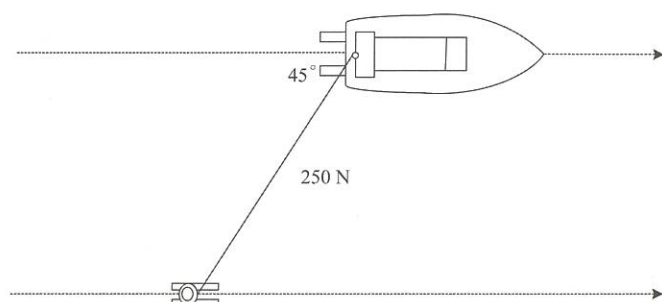


The Earth attracts it with a force of 480 N and the Moon attracts it with a force of 53.2 N when the asteroid is in position 1. The forces of attraction are 359 N and 13.1 N respectively when the asteroid is in position 2. Work out the net force on the asteroid in positions 1 and 2.

13. An archer fires an arrow with a velocity of  $45 \text{ m s}^{-1}$  at an angle of  $18^\circ$  to the horizontal. What is its velocity the horizontal direction?
14. A marathon runner is running at constant velocity. A spectator looking eastwards notices that the runner is moving northwards at  $1.40 \text{ m s}^{-1}$ . At the same time another spectator looking northwards notes that the runner is moving eastwards at  $1.10 \text{ m s}^{-1}$ . Determine the runner's actual velocity.
15. A two person bobsled crew push the sled, one from each side, at an angle of  $12^\circ$  to the track with a force 600 N each.
  - a) How much force do they apply to the sled parallel to the track?
  - b) What is the force with which each crew member pushes against the other?
16. A golfer hits a ball with a velocity of  $25 \text{ m s}^{-1}$  at an angle of  $35^\circ$  to the horizontal. Immediately after she hits the ball, at what rate does it rise and at what rate does it move towards the pin?
17. A roller skater starts down a slope inclined at  $25.0^\circ$  the horizontal. The skater is not significantly affected by frictional forces. Determine the skater's acceleration.
18. Rachael reaches a constant speed of  $35 \text{ km h}^{-1}$  while sliding down a water slide inclined at an angle of  $30.0^\circ$  to the horizontal. How long does she take to descend a vertical height of 12 m once she reaches this speed?

# Problem Solving and Calculations

## Set 1: Vector addition, subtraction and resolution

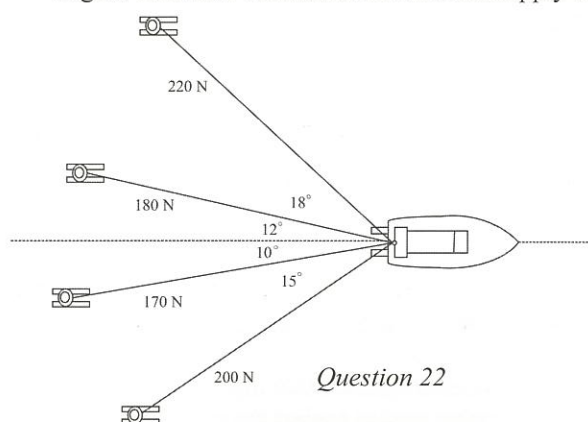


Question 19

19. A power boat tows a water skier. The skier is situated well to one side of the boat, but is travelling in the same direction as the boat. The tension in the rope is 25 N and the rope makes an angle of  $45^\circ$  to the direction of both boat and skier. Determine the minimum force the skier must apply to the water to maintain this path.

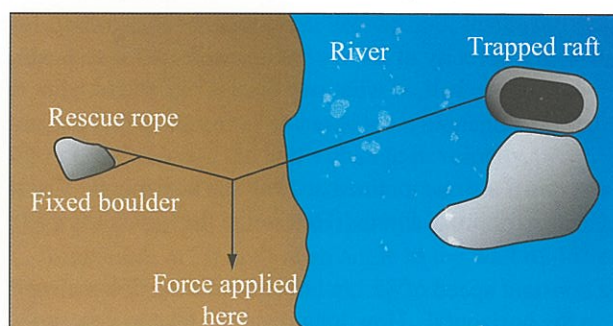
### Notes

20. During a football game a player kicks the ball eastward at a constant horizontal velocity of  $13 \text{ m s}^{-1}$  directly toward a team-mate 45 m away. There is a crosswind blowing from the north at  $8.5 \text{ m s}^{-1}$ .
- How much time does the team-mate have to get into position to catch the ball?
  - How far does the team-mate have to run in a southerly direction to catch the ball?
21. A truck driver supported a plank horizontally by its two ends. He then loaded bricks on its centre and found that the weight of 166 kg of bricks was just enough to break the plank. He wanted to use an identical plank as a ramp to load a full fuel drum of total mass 197 kg onto a truck.
- Explain why the driver could still use this plank to load the fuel drum.
  - What is the minimum angle to the horizontal at which he could still use the plank to load the drum?
22. A ski boat is towing four skiers. The diagram below shows the tow ropes' tensions and angles. Calculate the total force the skiers apply to the boat.



Question 22

23. A vector pull is a very simple method rescuers use to increase the force on a rescue rope during river rescues. Use a diagram with an appropriate scale to show that the force applied to the trapped raft is very much larger than the force applied by the rescuers.



Question 23